Spectral–Spatial Out-of-Distribution-Based Unsupervised Band Selection Method for Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral anomaly detection (HAD) aims to highlight the pixels that are different from the surrounding pixels without any prior information. However, as a hyperspectral image (HSI) tends to possess a huge data volume in the spectral domain, the dimension curse is inevitable in HAD. The unsupervised band selection (UBS) method is an effective tool to avoid the dimensionality curse in the HAD task. To obtain a more robust band subset without the help of any HAD detectors, we propose a spectral–spatial out-of-distribution (OOD)-based UBS method for HAD (HADUBS), which can acquire the optimal band subset in a more straightforward way. Our key observation is that the OOD term of pixels can reveal the differences and similarities of anomaly representation ability of different bands. Hence, we developed an OOD-based feature subspace representation module to obtain latent feature spaces with a better indication of the anomaly detection ability. Moreover, we introduced a UBS strategy called mutual information (MI)-based local outlier factor (MILOF) to significantly improve the discriminative ability of the selected band subset by investigating the locally sparse prior of anomalies. Extensive experimental results on five common HAD datasets demonstrate the superior performance of HADUBS. The source code will be made publicly available at https://github.com/duang33/HADUBS.**

***Index Terms*— Hyperspectral anomaly detection (HAD), hyperspectral image (HSI), latent feature spaces, local outlier factor, unsupervised band selection (UBS).**

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I. Introduction

HYPERSPECTRAL image (HSI) holds abundant spectral information with hundreds of spectral bands, which can record the spectral fingerprint for distinguishing different materials, and is therefore used for different processing tasks [1], [2], [3], [4]. Hyperspectral anomaly detection (HAD) [5], [6], as an important HSI processing task, has been successfully applied to different aspects, such as environmental monitoring [7], [8], medical diagnosis [9], and military warning [10] without the aid of ground-truth information. However, the performance of HAD has been seriously restricted by the redundant spectral bands. The detection accuracy exhibits an initial increase followed by a subsequent decrease as the number of bands increases. Therefore, it is necessary to perform dimensionality reduction (DR) before HAD processing.

Without the aid of any ground-truth information, the HAD aims to find those anomaly pixels that differ from their surrounding neighborhoods. Therefore, the unsupervised DR approach is more suitable for HAD. Generally, the unsupervised DR method can be categorized into two classes [11], unsupervised feature extraction and feature selection [12], i.e., unsupervised band selection (UBS). Although the former transforms the original data into a feature space and then obtains low-dimensional features, it discards the physical meaning of the raw data [13], [14]. In contrast, the UBS can select a representative band subset from the raw data to preserve the original physical meaning of HSI. Hence, we focus on the UBS methods for the HAD task.

Currently, most of the UBS methods are designed for the HSI classification task [15], [16]. By investigating the data characteristics of HSI to mitigate the limitation of ground-truth information, the UBS usually constructs some evaluation criteria to measure the significance of band subsets [17], and the potential optimal band subset can be acquired through an iterative or heuristics strategy [18], [19]. According to the strategy of acquiring band subsets, classification-task-oriented UBS methods can be divided into ranking-based [20], searching-based [21], and clustering-based methods [22]. Although they have achieved satisfactory results in the classification task, it is not realistic to directly apply these methods for the HAD as their main purpose is to find an effective band subset for the classification task. Although there are some

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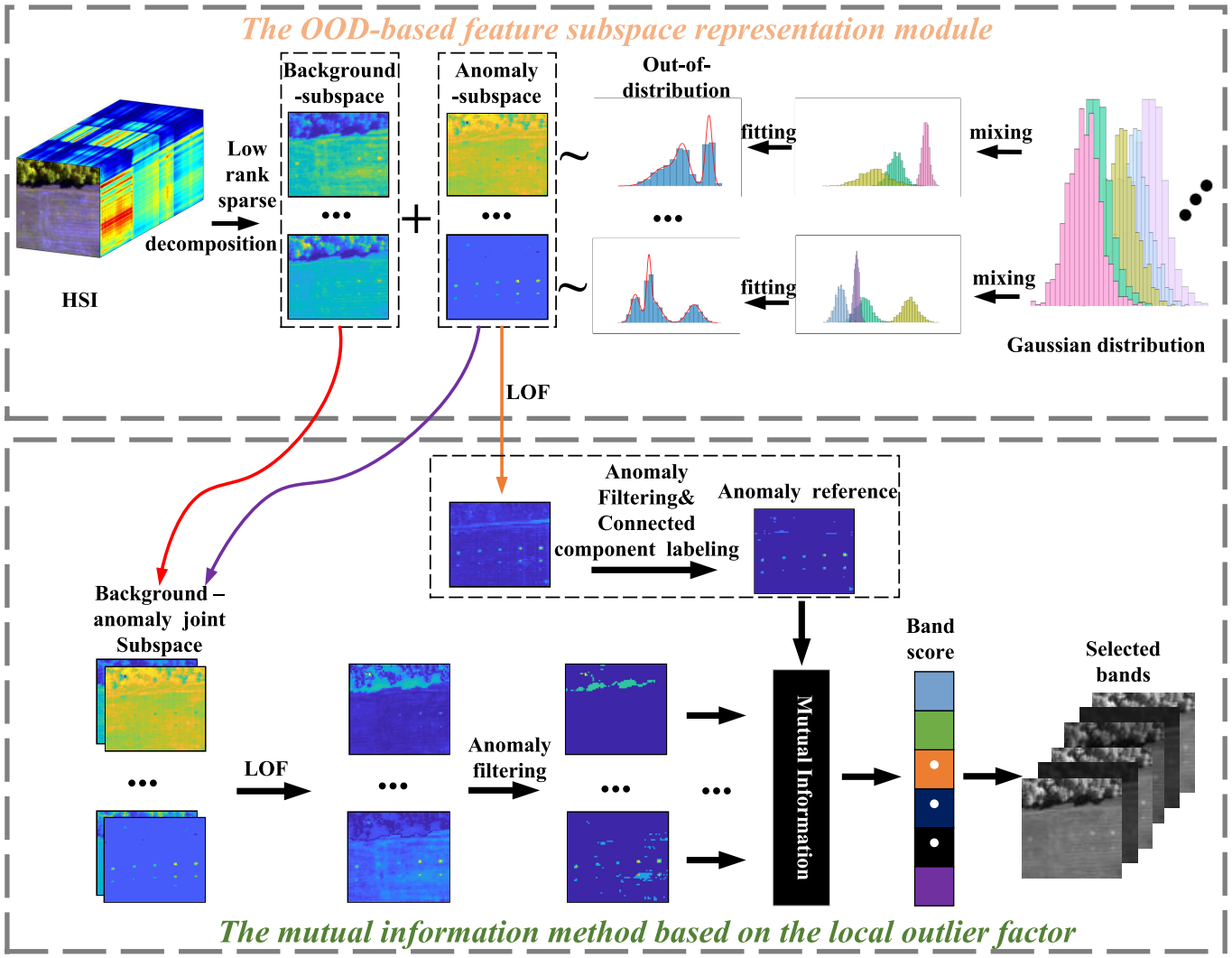


Fig. 1. Overall structure of our method.

HAD-oriented UBS methods nowadays, most of them rely on a detection result in advance, and then some band selection strategies are designed based on the results from the anomaly detector. While these methods have acquired some relatively good performance on the HAD task, their performance heavily relies on the anomaly detection process first, where each pixel is defined to be an anomaly or not by the detector. Therefore, these methods cannot obtain the optimal band subset for the HAD in a more straightforward way.

Compared with deciding whether the pixel is an anomaly or not by the detector, it is more reasonable to generate an out-of-distribution (OOD) term of pixels to evaluate the anomaly representation ability for each band [23]. As shown in Fig. 1, the OOD term of pixels is enormously different between band 157, which holds a poor anomaly, and band 16, which holds a clearer anomaly, while similar OOD exists between bands 157 and 158 due to their display of the similar anomaly representation ability. Accordingly, it can reveal the anomaly representation ability among bands. By generating the OOD of the whole HSI, it can adaptively model the data distribution for both the anomaly and background pixels, which enables

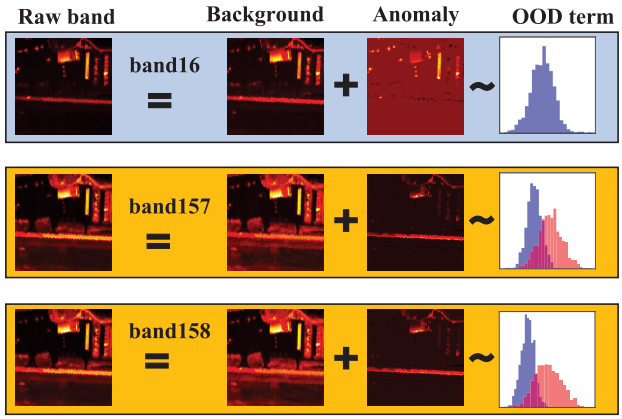


Fig. 2. Diagrammatic sketch of the OOD term for pixels in the urban3 dataset.

the OOD term to have a better ability to separate anomaly and background pixels without the aid of an anomaly detector. Therefore, we have proposed an HAD-oriented UBS method based on the OOD term of pixels in this article.

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TABLE I
DETAILS OF THE FIVE COMMON DATASETS FOR AD IN THE EXPERIMENT

Datasets	Spatial size	Spatial resolution	Number of bands	Spectral range	Anomaly pixels	Anomaly proportion	Sensors
forest	64×64	1.56m	169	400-2400nm	19	0.46%	HYDICE
urban	80×100	1.56m	175	400-2400nm	21	0.26%	HYDICE
urban3	100×100	3.5m	191	400-2500nm	52	0.52%	AVRIS
Hyperion	150×150	10m	149	350-2500nm	15	0.07%	Hyperion
San Diego airport	100×100	3.5m	189	400-2500nm	25	0.25%	AVRIS

To explore the OOD term on each spectral band, an OOD-based feature subspace representation module is desired to obtain a latent feature space for better indicating the anomaly detection ability of the band. Specifically, the OOD term of anomaly usually takes different properties on different bands as the reflectivity of anomalies is not the same on various wavelengths. Simultaneously, inspired by studies [24], [25], [26], [27] that demonstrate similarities between adjacent bands in HSI, it is evident that the representation of anomaly typically remains consistent across neighboring bands. To sum up, the OOD term on the spectral domain has the property of uniqueness and local correlation. As an effective tool for modeling data distribution, the Gaussian mixture model (GMM) [28] is a mixture model composed of multiple Gaussian distributions. However, it is difficult to determine the number of Gaussian distributions and fit the unique distribution characteristics of each band in HSI when applying the GMM model to the HSI processing task. By assuming each Gaussian distribution as a cluster, the Dirichlet process (DP) [29] can adaptively assign different pixels to different clusters according to the data property. In other words, it can achieve an autoclustering scheme without setting the number of clusters, and it makes certain data share parameters of the same Gaussian distribution (autoclustering), which provides an effective way to overcome the drawback of determining the number of Gaussian distributions. In this way, the OOD term among different bands can be adaptively modeled. For preserving the local correlation on the spectral domain, generating a similar Gaussian mixture distribution between adjacent bands is essential. Therefore, the Gaussian mixture distribution between adjacent bands should share some Gaussian distributions. For this reason, the higher level DP, namely, the hierarchical DP (HDP), can achieve this goal due to the property of autoclustering of DP mentioned above. To sum up, the HDP-based GMM (HDPGMM) can adaptively capture the uniqueness and local correlation of OOD terms among different bands, which enhances the ability to model the anomaly and background information.

As the anomaly usually takes up a small fraction of the whole HSI, the corresponding statistics as shown in Table I, the performance of HAD-oriented UBS methods will be easily affected by background pixels. Moreover, owing to the background usually containing some pixels that are different from their neighboring ones, these pixels will interfere the UBS process and some more discriminative bands will be ignored. Accordingly, it is essential to suppress the effect of background pixels with the aid of OOD terms during the UBS process. The anomaly has a sparse prior within the local region due to the anomaly typically occupying limited spatial extents. In other words, anomaly pixels are not densely connected. As shown in Fig. 3, anomaly pixels appear among the limited locally connected region of space. Therefore, it will be useful to use the local sparse prior of anomaly to avoid the interference resulting from the background pixels, and more discriminative bands can be obtained by acquiring the correlation of local sparse prior between different bands. Local outlier factor (LOF) [30] can compare the density difference between neighboring pixels and assign an anomaly degree to each pixel, which has the ability to effectively evaluate the local sparsity of anomalies. Accordingly, the local sparse prior of anomalies can be measured by the LOF. For obtaining the correlation between sparse prior among bands, mutual information (MI)-based LOF (MILOF) can help select more discriminative bands for anomaly because MI [31] is an effective tool to evaluate the correlation among different variables.

To sum up, this article proposed HAD-oriented UBS (HADUBS) methods based on the OOD term, which highlights a novel idea about HAD-oriented UBS methods by effectively describing the OOD term of anomalies among bands without the aid of any anomaly detector. Aiming to effectively reflect the uniqueness and local correlation of anomaly on the spectral dimension, the OOD-based feature subspace representation is designed to improve the representation ability of anomaly pixels. Based on the acquired OOD term, the MILOF is designed to evaluate the correlations among bands in the condition of acquiring local sparse prior, where an optimal band subset with strong anomaly separation capabilities can be efficiently chosen. The experimental results on several HAD datasets have proved the superiority of HADUBS.

The main contributions of this article are summarized as follows.

1) We make light on the HAD-oriented UBS based on the OOD term, which effectively avoids the reliance on the anomaly detectors, where an OOD-based feature subspace representation module is designed based on the HDPGMM, which can depict the uniqueness and local correlation of the OOD term on the spectral domain.
2) We propose a band selection strategy based on the MILOF, which significantly improves the discriminative ability of the selected band subset by investigating the locally sparse prior of anomalies.
3) We have compared our proposed method with current state-of-the-art methods on five commonly HAD datasets, and the experimental results have demonstrated the robustness and effectiveness of our proposed method.

The remainder structure of this article is constructed as follows. Section II reviews the related UBS work. Section III presents the details of HADUBS, including the OOD-based feature subspace representation method for OOD terms and the MILOF. Afterward, Section IV reports the experiment

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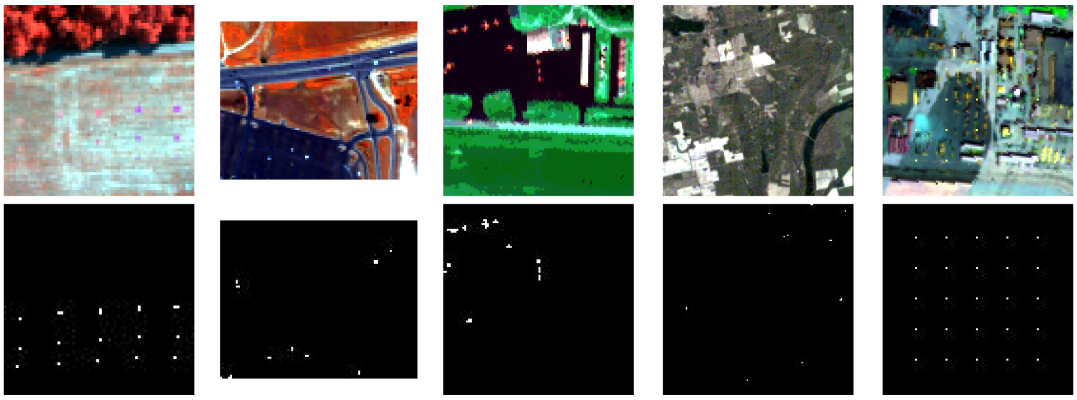


Fig. 3. First line represents the pseudocolor map and the second line represents the truth map, from left to right, forest, urban, urban3, Hyperion, and San Diego airport.

results and discussions. Finally, the conclusion is given in Section V.

II. Related Work

In this section, we will review the literature involving HAD, common UBS methods, and HAD-oriented UBS methods in turn.

A. HAD

As a classic statistical method, the Reed–Xiaoli (RX) method has been widely applied to the HAD task. It assumes that the background follows a multivariate Gaussian distribution, and the detection map can be acquired by computing the Mahalanobis distance between each pixel and its neighboring pixels. However, the background may also contain some anomaly pixels. To alleviate the dilemma, some improved RX-based methods have been developed to better model the background information, such as the SRX-local [32] and the SRX-segmented [33]. Although these methods have obtained certain success, they usually cannot obtain a better result with a complex background as the background information of an HSI may not always follow the aforementioned Gaussian distribution assumption.

To improve HAD performance, some work have attempted to use the prior information of HSI to better model the background information. This kind of methods usually considers that background has a low-rank characteristic and anomaly has a sparse property, and some researchers have used the low-rank and sparse matrix decomposition (LSDM) [34], [35], [36], [37] technology into the HAD task. However, the above-mentioned traditional LSDM model assumes that sparse component follows a single distribution, which is not reasonable due to the tendency to a mixed pixel that has weak anomaly combined with many kinds of noise. In this respect, Ding et al. [28] have proposed an LSDM-mixture of Gaussian (LSDM-MoG), which adopts an MoG to fit the sparse item and achieve better results. Nevertheless, the presence of correlation and difference in anomaly among bands, and the number of Gaussian distributions of sparse components in each band should be different, which challenges the adaptive separation between the background and anomaly.

B. UBS

According to the basic operating principle of UBS, UBS can be divided into three categories: search-based, clustered-based, and ranking-based methods. As for the search-based method [17], it attempts to search the all-band combination as possible. Different from it, the clustered-based methods [38] obtain the band subset by selecting the representative band in each group. The ranking-based method [39] adopts a more direct way, whose basic principle is to endow a score for each band. Afterward, select the band that holds the highest score. For each category, a detailed introduction is as follows.

The searching-based methods transform the UBS into an optimal problem by predefined some criterion function to find the best band subsets. Here, the criterion function usually was derived from some HSI prior information. Based on the prior information of neighboring bands having a stronger correlation, Chang et al. [21] proposed the optimal neighborhood reconstruction (ONR) to find the band subsets by the linear reconstruction of the whole HSI.

Unlike search-based methods, clustering-based methods separate the bands into several clusters and select the most discriminative band in each cluster. To balance the computational cost and UBS performance, an optimal clustering framework (OCF) [22] has been proposed based on dynamic programming, which can obtain the optimal clustering result and select the most representative one in each cluster with different metrics, such as the E-FDPC [40], information entropy (IE) [41], and maximum-variance PCA-based scoring criterion. As noisy bands tend to become a single cluster, most clustering-based methods are very sensitive to noise, and Zhang et al. [38] have designed the adaptive distance-based band hierarchy (ADBH) method, which designs a novel measurement by integrating Euclidean distance with the cluster density to improve robustness to noise bands. Subsequently, a novel hyperbolic clustering-based band hierarchy (HCBH) has been proposed [39] to fully represent the underlying spectral structure and improve the performance of UBS.

The ranking-based method aims to evaluate each band according to some designed strategy and then selects the top bands as the optimal band subsets. Recently, Sun et al. [40] have introduced the enhanced fast density-peak-based

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clustering (E-FDPC) by jointly considering the local density and the intracluster distance of each band. To sum up, ranking-based methods are very intuitive compared with the searching-based and clustering-based methods, and the selected bands could be from certain spectrum range. Different from the classification task to obtain band subset uniformly, it is more favorable to select bands from a spectrum range for the HAD task as the difference between anomaly and background may not exist in the whole spectrum. Hence, the ranking-based methods are the main focus of this article.

C. HAD-Oriented UBS

Currently, there are not much UBS methods designed for HAD tasks, and those UBS methods designed for the classification task may not select the optimal band subset for the HAD task.

The early work of the HAD-oriented UBS method can be traced back to the recursive SAM-based (RSAM) methods [42], which select the band subset iteratively according to the separability of background and anomaly pixels until a predefined number of bands. He et al. [43] have proposed a sequential band subset selection (SQ-BSS) algorithm or successive BSS (SC-BSS) algorithm that adapts the endmembers' extraction algorithm to a UBS task by treating each band as a sample vector. Moreover, Wang et al. [44] have introduced a physical constraint variational autoencoder (VAE) to UBS, and the selected band subset contains discriminative features between anomalies and background [45]. Aiming to introduce some prior information in HAD-oriented UBS methods, Shang et al. have proposed the residual-driven band selection (RDBS) method, which uses the information from the RX as a band selection prior [46].

However, the above methods rely heavily on the performance of an HAD detector, which raises a conflict issue for the UBS. If the HAD detector obtains an effective performance, it is not necessary to perform the UBS anymore. Therefore, we focus more on the HAD-oriented UBS method without the aid of detector, which selects the desired band subset by investigating the distribution of background and anomaly.

III. Methodology

Motivated by the strong anomaly representation ability of OOD term for each band, we have proposed an HAD-oriented UBS method (HADUBS) based on the spectral–spatial OOD information. In this section, the HADUBS will be introduced in detail, including the OOD-based feature subspace representation method and the band selection strategy with the aid of MI and local outlier factor (LOF).

A. OOD-Based Feature Subspace Representation

For the HAD task, it is more desirable to obtain a band subset with a strong ability of distinguishing the anomaly and background. As our proposed method does not rely on any detector, we focus more on investigating the data distribution of each band, which contains the discriminative information for better characterizing the anomaly and background. As the difference between the anomaly and background is usually not clear enough, most HAD methods have attempted to project the original data into a feature subspace. Similarly, we have designed a OOD-based feature subspace representation module. Unlike the original data, the generated OOD-based feature subspace can better observe the anomalies in the bands and thus characterize the anomaly of different bands. To simultaneously consider the correlation among consecutive bands and the diversity between different bands, we have obtained the OOD-based feature of each band according to the HDPGMM [29], which can better model the complex distribution of HSI data.

The overall structure is depicted in Fig. 1. Let $\mathbf{Y} \in R^{N\times L}$ be an HSI in which N represents the number of pixels and L indicates the number of bands. Firstly, the input HSI $\mathbf{Y}$ is decomposed into the background subspace $\mathbf{B} \in R^{N\times L}$ and the sparse subspace $\mathbf{A} \in R^{N\times L}$ mixed with the noise and anomaly with the aid of LSDM model. The process can be formulated as follows:

$$\mathbf{Y} = \mathbf{B} + \mathbf{A}. \tag{1}$$

Based on the assumption that the background of HSI possesses a low-rank property, the background space $\mathbf{B}$ can be viewed as the product between $\mathbf{U} \in R^{N\times l}$ and $\mathbf{V} \in R^{L\times l}$ of two smaller matrices. The corresponding formula is defined as follows:

$$\mathbf{B} = \mathbf{U}\mathbf{V}^T = \sum_{r=1}^{l} \mathbf{u}_{.r}\mathbf{v}_{.r}^T. \tag{2}$$

Noted that $l <= L$ is a predefined parameter, while $\mathbf{u}_{.r}$ and $\mathbf{v}_{.r}$ can be acquired from two multivariate Gaussian distributions

$$\mathbf{u}_{.r} \sim \mathcal{N}\left(\mathbf{0}, \lambda_r^{-1}\mathbf{I}_N\right), \mathbf{v}_{.r} \sim \mathcal{N}\left(\mathbf{0}, \lambda_r^{-1}\mathbf{I}_L\right) \tag{3}$$

where $\mathbf{I}_N$ and $\mathbf{I}_L$, respectively, represent $N \times N$ and $L \times L$ identity matrices. In addition, the parameter λ_r follows Gamma distribution with parameters predefined p_0 and q_0.

To adaptively separate the OOD and background distributions in HSI, it is insufficient to focus solely on the background subspace; the anomaly subspace must also be considered and specifically designed. In this work, the corresponding characteristic is also reflected in the anomaly as the presence of distinctness and correlations among bands. Through HDPGMM, the distinction among bands can be modeled with the form of different Gaussian distribution in the first level of HDPGMM, i.e., completing the process of autoclustering for each pixel in each band, and the correlation between adjacent bands can be represented by sharing the same Gaussian distribution in the second level of HDPGMM.

Specifically, the ith row and jth column of the anomaly subspace follow the below Gaussian mixture distribution:

$$a_{ij} \sim \sum_{k=1}^{K} \pi_{jk}\mathcal{N}\left(0, \xi_k^{-1}\right) \tag{4}$$

where K represents the total number of Gaussian mixture distributions, and the above-mentioned formula can be transformed into the following form:

$$a_{ij} \sim \mathcal{N}\left(0, \xi_{z_{ij}}^{-1}\right), z_{ij} \sim \text{Multi}\left(\boldsymbol{\pi}_j\right) \tag{5}$$

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where ξ_k follows a Gamma distribution with parameters a_0 and b_0, which are predefined defaults. The $\boldsymbol{\pi}_j$ is a vector with a length of K and $1 \leq z_{ij} \leq K$. It follows a hierarchical Dirichlet distribution, i.e.,

$$\boldsymbol{\pi}_j \sim \mathrm{Dir}(\alpha_j \boldsymbol{\beta}), \boldsymbol{\beta} \sim \mathrm{Dir}(\gamma/K, \gamma/K, \ldots, \gamma/K). \quad (6)$$

For the purpose of better understanding the hierarchical Dirichlet distribution (Dir(.)), the notations involved in the DP are introduced as follows. DP is a stochastic process, which can be defined by a concentration parameter γ and a base distribution H, i.e., DP(γ, H). $\boldsymbol{\theta}$ is a measurable set that can be divided into random finite partitions $\{A_1, A_2, \ldots, A_r\}$. DP($\gamma$, H) that follow the below conditions is the unique distribution over $\boldsymbol{\theta}$:

$$G \sim \mathrm{DP}(\gamma, H) \quad (7)$$

$$(G(A_1), G(A_2), \ldots, G(A_r)) \sim \mathrm{Dir}(\gamma H(A_1), \gamma H(A_2), \ldots, \gamma H(A_r)). \quad (8)$$

There are two essential properties in DP. First, any sample G drawn from the Dir(γ, H) is a discrete distribution over $\boldsymbol{\theta}$. Second, the sample $\theta_i \sim G$ drawn from the discrete distribution G is repeatable, i.e., $\theta_i = \theta_j$, where $i \neq j$, $i = 1, 2, \ldots i, \ldots, N$. According to the two properties, the HSI pixels that contain similar spectral characteristics will share the same θ_i, and pixels of various characteristics will be placed on different θ_i, where each θ_i can be considered as a set composed of parameter mean and variance in Gaussian distribution. Accordingly, the GMM model can perform automatic clustering with the aid of DP.

Aiming to sample the discrete distribution G from the DP, Shang et al. [47] proposed the stick-breaking construction which can be formulated as follows:

$$\pi_i = \pi_i' \prod_{j=1}^{i-1} (1 - \pi_j'), G = \sum_{i=1}^{\infty} \pi_i \delta_{\eta_i^*} \quad (9)$$

where $\pi_i' \sim \mathrm{Beta}(1, \gamma)$ and $\eta_i^* \sim H$ are two collections that are composed of infinite independent random variables. The $\delta_{\eta_i^*}$ stands for the Dirac Delta function located at η_i^*. Note that π_i can be viewed as the mixing coefficient so that stick-breaking construction is able to naturally depict the properties of clustering of DP. Simultaneously, the parameter γ can regulate the discrete level of π_i. The smaller the γ, the more discrete the π_i, and the fewer the number of Gaussian components of GMM. Therefore, it adaptively tuned up the number of Gaussian components of GMM.

The aforementioned process, which adopts only one DP into GMM, failed to depict the anomaly association among bands. To this end, the two DPs are used by drawing from the first DP, which enables some same Gaussian components to exist in various bands. As shown in the upper right of Fig. 1, the Gaussian distribution of the same color occurs in different band anomaly subspaces. The HDP can be formulated as $G \sim \mathrm{DP}(\gamma, H), G_j \sim \mathrm{DP}(\alpha_j, G)$, where G_j represents the distribution G belongs to the jth band group with being automatically determined in accordance with DP.

To improve the flexibility of the proposed model, the model can be reformulated as follows by assuming $K \to \infty$:

$$a_{ij} \sim \mathcal{N}(0, \Psi_{ij}^{-1}), \Psi_{ij} \sim G_j, G_j \sim \mathrm{DP}(\alpha_j, G), G \sim \mathrm{DP}(\gamma, H)$$
$$H = \mathrm{Gam}(a_0, b_0). \quad (10)$$

To be more specific, the model can be reformulated as below by introducing two coupled latent variables z_{ij} and c_{jt} with the aid of mentioned stick-breaking construction

$$c_{jt} \sim \mathrm{Multi}(\boldsymbol{\beta}), \beta_k = \beta_k' \prod_{l=1}^{k-1} (1 - \beta_l'), \beta_k' \sim \mathrm{Beta}(1, \gamma)$$
$$\xi_k \sim \mathrm{Gam}(a_0, b_0), z_{ij} \sim \mathrm{Multi}(\boldsymbol{\pi_j}), \pi_{jt} = \pi_{jt}' \prod_{t=1}^{s=1} (1 - \pi_{js}')$$
$$\pi_{jt}' \sim \mathrm{Beta}(1, \alpha_j), a_{ij} \sim \mathcal{N}\left(0, \xi_{c_{j,z_{ij}}}^{-1}\right). \quad (11)$$

It should be noted that ξ existing in (12) has the following relationship with Ψ involved in (10):

$$\xi_{c_{j,z_{ij}}} = \Psi_{ij}. \quad (12)$$

Aiming to further improve the model adaptability with respect to adjusting the number of GMM, we have defined α and γ involved in (10) to follow the below condition:

$$\gamma \sim \mathrm{Gam}(c_0, d_0), \alpha_j \sim \mathrm{Gam}(e_0, f_0). \quad (13)$$

It is worth mentioning that $a_0, b_0, c_0, d_0, e_0, f_0, p_0, q_0$ are the hyperparameters of the prior distributions. They were set to a small value to make prior distributions noninformative which failed to be affected by the variety of hyperparameters. Moreover, the truncation strategy [48] was used to improve the solvability, i.e., the number of GMM in the first level and the second level was set to be K and T (T and K enough large), respectively, in the condition of T $\leq$ K. Afterward, z_{ij} involved in (12) whose range can be adjusted to $1 \leq z_{ij} \leq$ T. It refers to the index of the Gaussian component from a_{ij} generated in the second-level GMM of band j. The c_{jt} ($1 \leq c_{jt} \leq$ K, $1 \leq t \leq$ T) stands for the index of the Gaussian component of GMM in the first level. As the anomaly pixels of anomaly targets generally exist in a Gaussian distribution and the noise (it can also be seen as a type of anomaly pixel) was depicted in the remaining Gaussian distribution, it is essential to separate the optimal Gaussian distribution as the anomaly subspace with better anomaly representation ability in each band. That is to say, the sparse subspace $\mathbf{A}$ has not only been considered but the corresponding z_{ij} with regard to the vary of Gaussian distribution needs to be derived.

Finally, the full Bayesian model for characterizing the OOD can be formulated as follows by combining the above equation:

$$p(\boldsymbol{U}, \boldsymbol{V}, \boldsymbol{\xi}, \boldsymbol{C}, \boldsymbol{Z}, \boldsymbol{\beta}, \boldsymbol{\pi}, \lambda, \boldsymbol{\alpha}, \gamma | \boldsymbol{Y}). \quad (14)$$

To optimize the Bayesian problem, the variational Bayes have been adopted to estimate the posterior distribution. It assumes that variable set Θ consisted of every variable in the full Bayesian model. The estimation $q(\Theta_k)$ of the posterior distribution of the kth variable can be formulated as below

$$\log q(\Theta_k) = E(\log p(\Theta))_{\Theta/\Theta_k} + \mathrm{const} \quad (15)$$

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where E(.) stands for the mathematical expectations' operator. Θ/Θ_k represents all the variables except Θ_k in variable set Θ. All the variables can be inferred by the above way until the low-rank matrix $\mathbf{B}$ converges or reaches the maximum iteration. The corresponding infer stage can be described in the following pseudocode.

Algorithm 1 Pseudocode for OOD-Based Feature Subspace Representation Stage

Input: HSI data $\mathbf{Y} \in R^{N\times L}$; the rank l involved the background matrix $\mathbf{B}$; the number K of the Gaussian distribution in the first level of HDP; the number T of the Gaussian distribution in the second level of HDP, and the maximum number of iterations $Iter$.

Output: matrix $\mathbf{U}$, $\mathbf{V}$, and the $\mathbf{Z}$.

while *the matrix* $\boldsymbol{B}$ *not converging or the number of iterations does not reach the* $Iter$ **do**
  Update the variables of the anomaly component: $\boldsymbol{\xi}$, $\boldsymbol{C}$, $\boldsymbol{Z}$, $\boldsymbol{\pi}$, and $\boldsymbol{\beta}$.
  Update the variables of the concentration component: $\boldsymbol{\alpha}$, and $\boldsymbol{\gamma}$.
  Update the variables of the LSDM components: $\boldsymbol{U}$, $\boldsymbol{V}$, and λ.
The specific closed-form solution of the inferred formula can be shown in the attached file.

$\mathbf{Z}$ can be obtained by the above derivation, which records the index of the Gaussian distribution of different pixels. Accordingly, the matrix $\mathbf{A}$ can be obtained by the difference operation between $\mathbf{Y}$ and $\mathbf{UV}$, i.e., $\mathbf{A} = \mathbf{Y} - \mathbf{U}\,\mathbf{V}^T$. Generally speaking, the anomalies' subspace of the jth band (a'_j) with better anomaly representation ability possesses the greater pixel value gap, which can be formulated as

$$\mathbf{a}'_j = \underset{\mathbf{a}_{.j}\circ\mathbf{z}_{.j}}{\arg\max}\left(\max\left(\mathbf{a}_{.j}\circ\mathbf{z}_{.j}\right) - \min\left(\mathbf{a}_{.j}\circ\mathbf{z}_{.j}\right)\right) \tag{16}$$

where $\circ$ represents the Hadamard product.

B. Mutual Information Method Based on the Local Outlier Factor

Through the above representation method, the background distribution and the OOD can be adaptively separated, and the background and anomaly subspace of each band can be obtained. With the assistance of two subspaces, we can evaluate the ability of background suppression and anomaly detection for various bands. However, it is not sufficient for an HADUBS task due to the presence of more background pixels is unfavorable to the HAD task. Moreover, the sparse attribution of anomaly targets within local space is lacking in consideration. Hence, we have proposed the MILOF method to explore the correlation of local sparse prior between bands in the condition of alleviating the effects of background pixels.

First, as the anomaly subspace considers the anomaly that exist in each band, its overall combination which involves the anomaly of the overall band can be viewed as a better anomaly reference foundation compared with the sole anomaly subspace. To explore the anomaly reference diagram from the overall combination of anomaly subspace, the definition of the HAD task needs to be sighted in advance. The HAD task usually refers to highlighting the pixels that have significant differences in spectrum compared with the surrounding pixels. Local outlier factor (LOF), as a common anomaly detection method, has been successfully applied to the color image yet the HSI is rare. It enables each pixel to get an anomaly score by comparing the difference with the surrounding pixels which is consistent with the definition of the HAD task. Therefore, LOF can entrust the higher score to the anomaly pixel so as to highlight the anomaly pixel.

Some notions are given as follows. Let $\mathbf{A}' = \{\mathbf{a}'_1, \mathbf{a}'_2, \ldots, \mathbf{a}'_i, \ldots, \mathbf{a}'_L\} \in R^{N\times L}$ represent the overall anomaly subspace combination. The kth distance of pixel O can be defined by $d_k(O)$, which means the distance of the kth closest pixel to pixel O. Let the reachability distance between pixel P and O be equal to $\text{reach-dist}_k(P, O) = \max\{d_k(O), d(P, O)\}$, where $d(P, O)$ indicates the Euclidean distance between pixel P and O. Afterward, local reachability density of pixel P can be denoted by $\text{lrd}_k(P) = (\frac{1}{k}\sum_{o\in N(P,k)} \text{reach-dist}_k(P, o))^{-1}$, where $N(P, k)$ is the set of k nearest neighbors of pixel P. According to the mentioned definition, the LOF of pixel P can be formulated as follows:

$$\text{LOF}_k(P) = \frac{1}{k}\sum_{o\in N(P,k)} \frac{\text{lrd}_k(o)}{\text{lrd}_k(P)}. \tag{17}$$

By computing the LOF of each pixel in $\mathbf{A}'$, the anomaly reference diagram can be obtained and defined as $\mathbf{A}^r \in R^{W\times H}$, where $W \times H = N$. However, as the more background pixels, the task is mainly affected by the background pixels. To avoid this unreasonable situation, it is necessary to screen the anomaly pixels in accordance with the anomaly score and the local sparsity of anomaly targets. Specifically, the pixel that possesses the higher score tends to be more like an anomaly pixel. Therefore, we adopt anomaly filtering processing to $\mathbf{A}^r$ and receive the $\mathbf{A}^t \in R^{W\times H}$, where let the pixel equal to zero in the condition of its value less than the top ε % pixel in $\mathbf{A}^r$. It should be noted that ε has been set as a fixed constant and its robustness is validated in the subsequent experiment. Although most interference from the background pixels has been eliminated by the above means, some still exist due to the local sparsity of anomaly targets is not taken into account.

Connected component labeling can obtain anomaly areas of different sizes after anomaly filtering, which benefits the measurement of sparsity level of anomaly areas. The two-pass method [49], which needs to be traversed twice for each pixel in a binary image, is a common connected component labeling method. Here, the first traversing is to assign a marker for each pixel and then record which connected domain each marker belongs to. It can be signified as "FP(.)." The second traversing is to mark each pixel as belonging to a connected component. It can be signified as "SP(.)." To this end, the process of two-pass can be formulated as follows:

$$\mathbf{A}^{t,\text{label}} = SP\left(FP\left(\mathbf{A}^t\right)\right). \tag{18}$$

We define the acquired connected region set as $C = \{c_1, c_2, \ldots, c_i, \ldots\}$ in $\mathbf{A}^{t,\text{label}}$. It should be noted that c_i stands

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for the ith connected region which has the property of the number of pixels. For the sake of carrying out the sparse property of anomaly targets, we attempt to reserve c_i which possesses a smaller number of pixels and discard the pixels existing in the residual set, which can be formulated as follows:

$$c_i = \begin{cases} c_i, & \text{num}(c_i) < \phi \\ 0, & \text{else} \end{cases} \tag{19}$$

where $\text{num}(c_i)$ represents the number of pixels in c_i, and ϕ stands for hyperparameters which also be set as a fixed value in our method. The corresponding experiment has been validated in Section IV-C. Afterward, the pixels that were only saved in some c_i emerge in the last anomaly reference diagram which can be defined as $\mathbf{A}^l$. Accordingly, it can be as a criterion to measure the difference in local sparsity of anomaly targets among bands.

To obtain more differentiation in different bands, various bands adopt the combination of respective background and anomaly subspace, whose combination of the ith band can be indicated as CBA_i, The LOF method was repeatedly applied in CBA_i and acquired the anomaly degree diagram. Afterward, aiming to delay the main background pixels' impacts and retain the different sparsity of anomaly targets among bands, anomaly filtering was conducted in the anomaly degree diagram of each band, which is different from the production of the anomaly reference diagram with connected component labeling. The anomaly degree diagram of the ith band after anomaly filtering can be expressed as CBA_i^l.

Subsequently, to consider the relationships of local sparsity of anomaly targets among bands and assign priority to different bands, we adopt the MI method to achieve this goal. This is caused by the MI which can measure the degree of correlation between variables. In this way, the distinct between variables CBA_i^l and $\mathbf{A}^l$, i.e., how close the anomaly of the ith band is to anomaly reference diagram with closer to reality, can be measured by the MI. The corresponding formulas can be indicated below

$$\begin{aligned} \text{MI}\left(\text{CBA}_i^l, A^l\right) &= H\left(\text{CBA}_i^l\right) + H\left(\mathbf{A}^l\right) - H\left(\text{CBA}_i^l, \mathbf{A}^l\right) \\ H\left(\text{CBA}_i^l\right) &= \sum_{m=0}^{Q-1} \left(-P_{\text{CBA}_i^l}(m)\right) \log P_{\text{CBA}_i^l}(m) \\ H\left(\text{CBA}_i^l, \mathbf{A}^l\right) &= \sum_{m=0,n=0}^{Q-1} \left(-P_{\text{CBA}_i^l, \mathbf{A}^l}(m,n)\right) \log P_{\text{CBA}_i^l, \mathbf{A}^l}(m,n) \end{aligned} \tag{20}$$

where Q is the number of different grayscale values within the image. It is set to be 255. $P_{\text{CBA}_i^l}(m)$ stands for the probability of grayscale equal to m in the image CBA_i^l. $P_{\text{CBA}_m^l, \mathbf{A}^l}(m,n)$ represents the probability that the grayscale is equal to m in image CBA_i^l and the grayscale is equal to n in $\mathbf{A}^l$ in the condition of the pixel occurring in the same position between image CBA_i^l with image $\mathbf{A}^l$.

Therefore, the priority of each band can be obtained by the above process, i.e., $\text{MI}(\text{CBA}_1^l, \mathbf{A}^l), \text{MI}(\text{CBA}_2^l, \mathbf{A}^l), \ldots, \text{MI}(\text{CBA}_L^l, \mathbf{A}^l)$. To clearly explain the procedure of HADUBS, the corresponding pseudocode can be summarized in Algorithm 2. To sum up, the MILOF identifies the local sparse prior for anomaly objects and establishes a relationship between the anomaly reference and the feature subspace of each spectral band. Following this, it assigns varying priorities to the different bands based on their significance in the HAD process.

Algorithm 2 Pseudocode for the HADUBS

Input: HSI data $\mathbf{Y} \in R^{N \times L}$

Output: The sequence of band priority: $\text{MI}(CBA_1^l, \mathbf{A}^l), \text{MI}(CBA_2^l, \mathbf{A}^l), \ldots, \text{MI}(CBA_L^l, \mathbf{A}^l)$.

1: According to the OOD-based feature subspace representation module, obtaining the anomaly subspace $\mathbf{a}_i^{'}$ and the combination of respective background and anomaly subspace CBA_i of each band, taking the ith band as an example. Therefore, the overall anomaly subspace combination can be represented as $\mathbf{A}^{'} = \{\mathbf{a}_1^{'}, \mathbf{a}_2^{'}, \ldots, \mathbf{a}_i^{'}, \ldots, \mathbf{a}_L^{'}\} \in R^{N \times L}$.
2: The LOF is performed for $\mathbf{A}^{'}$ and CBA_i before obtaining the grayscale image $\mathbf{A}^r$ and CBA_i^r, respectively.
3: The anomaly filtering and the two-pass (connected component labeling) method are performed sequentially for $\mathbf{A}^r$ and acquire the anomaly reference diagram $\mathbf{A}^l$.
4: The anomaly filtering is applied for CBA_i^r of each band and achieve CBA_i^l.
5: The MI is carried out between CBA_i^l with $\mathbf{A}^l$ of each band and obtains the priority of each band $\text{MI}(CBA_i^l, \mathbf{A}^l)$.

IV. Experiments and Analysis

In this section, we have conducted the comparison experiment with eight existing UBS methods involving HAD-oriented UBS in five common HAD datasets to demonstrate the effectiveness and robustness of our proposed method.

A. Datasets' Description

In actual HAD tasks, abnormal targets are usually a minority, so we choose the five datasets with a small percentage of anomaly which are forest, urban, urban3, Hyperion, and the San Diego airport to validate the performance of the proposed method, whose details are summarized in Table I.

1) Forest: Collected by the HYDICE sensor, this dataset has 169 spectral bands with a spectral range from 400 to 2400 nm and a spatial 64 × 64 pixels. The dataset exhibits 19 anomaly pixels originating from 15 panels in various scales, which account for 0.46% of the total number of pixels.

2) Urban: With 175 spectral bands in a wavelength range of 400–2400 nm, this dataset was captured by the HYDICE sensor. There exist two roofs and cars with 21 pixels to be detected, constituting 0.26% of the total number of pixels.

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3) Urban3: Captured by AVRIS sensor and with 191 bands ranging from 400 to 2500 nm, this dataset has a spatial extent of 100 × 100, and the anomalies come from buildings with 52 anomaly pixels, which occupy 0.52% of the total number of pixels.

4) Hyperion: This dataset comprises 150 × 150 pixels and contains 149 bands, ranging from 350 to 2500 nm by Hyperion sensor. The Storage silo and the roof exhibited anomaly characteristics, with 15 anomaly pixels accounting for 0.07% of the total number of pixels.

5) San Diego Airport: This dataset has been manufactured in work [5]. The background and anomaly spectrum were captured by AVRIS sensor within the spectral range from 400 to 2500 nm, which involved San Diego airport and aircraft, with 25 anomaly pixels accounting for 0.25% of the total number of pixels. They are regularly arranged in 5 × 5.

The pseudocolor and ground-truth maps of these scenarios are shown sequentially in Fig. 3.

B. Experimental Setup

1) Comparison Methods: As there are not much HAD-oriented UBS methods, we have compared the proposed method with seven advanced UBS methods in both the classification and HAD tasks, including ONR [21], UBS, EPDPC [40], TRC_OC_FDPC [22], HCBH [39], ADBH [38], and the HAD-oriented UBS method RDBSMaxSR [46]. ONR is a searching-based method, TRC_OC_FDPC, HCBH, and ADBH are clustering-based methods, while the EFDPC and RDBSMaxSR are ranking-based methods. These parameters of compared methods were set as the recommended parameters in accordance with the original literature. RX detector, as a common HAD method, can be widely used for different HAD scenes without any prior knowledge. Therefore, it was uniformly adopted for estimating the HAD performance of the selected BS subset in the experiment section. It should be noted that $a_0, b_0, c_0, d_0, e_0, f_0, p_0, q_0$ was set as 10^{-6} caused by such in a noninformative manner that leads to results being stable, and we set T and K to 7 and 15, respectively, due to the Gaussian distribution commonly less than such number in the real situation. Let rank l involved in the background matrix B be equal to 3, and $Iter$ is set as 300.

To validate the effectiveness of the OOD-based feature subspace representation method, the generated OOD terms have been evaluated by attaching the existing five detectors which are classical RX and the LSDM-related ones due to the following two points. First, the assumption of RX is that the data follow a single distribution, which is consistent with that of OOD is a single distribution. Second, the representation method was originally based on the LSDM approach. The five existing methods are RX, RPCA-RX [50], LRASR [36], LSMAD [37], and LSDM-MoG-RX [28], and the corresponding parameters are set as suggested in the literature.

2) Evaluation Metrics: To measure the performance of the proposed method, we have adopted both qualitative and quantitative evaluation metrics. Accordingly, the anomaly map, the area under curve (AUC), and the box plot after detecting different band subsets have been conducted in this article. Anomaly maps qualitatively offer insights into how well the method detects anomalies while suppressing background. The curves of AUC quantitatively can observe the robustness of different UBS methods. The anomaly background box plot is able to visualize the degree of anomaly prominence or suppression of background between different UBS methods.

It should be noted that all the experiments are performed on an Ubuntu 22.04 system with CPU 12900K and memory 32G, and the codes of both the compared and our methods are implemented on the MATLAB 2022b.

C. Parameter Determination

In the HADUBS method, the hyperparameter ε involved anomaly filtering, and the ϕ involved connected component labeling is essential for robustness. Therefore, an extensive experiment has been conducted by fixing ε and then adjusting ϕ in three datasets including forest, urban, and San Diego airport. The left of Fig. 4 shows the situations that empirically fix ε as 96 and adjust ϕ from 5 to 30 by taking 5 as the interval, where the average accuracy of the band subset is the average of nine band subsets varying from 10 to 50 at intervals of 5. It should be noted that the stable trend after $\phi =$ 25 occurs in three datasets. Accordingly, let ϕ equals 25 and adjust ε continuously. In the first two datasets, the trend of curves before $\varepsilon = 98$ is relatively stable; however, in the San Diego airport, a significant mutation occurred when $\varepsilon = 98$. Thereby, as long as ε is less than or equal to 98, we will achieve a relatively robust performance. We simply set ε as 96. To sum up, we have set ε and ϕ to be 96 and 25 in the experiments, respectively.

D. Comparison Results

In this section, a unified analysis of the detection performance of different band subsets from various UBS methods has been given. First, the AUC of the band subsets by the RX detector is shown in Fig. 5. These experiments record the AUC of nine band subsets whose size varied from 10 to 50 with five intervals.

In Fig. 5(a), HADUBS outperforms the other UBS methods by a tiny margin when the size of the band subsets is greater than 25 on the forest, with stable and high AUC against all the compared methods. When the size is less than 25, there is no advantage in our method; instead, the EFDPC method and RDBSMaxSR are highly competitive, probably due to the robustness of these methods to noise, which makes them more suitable for noisier datasets like forest.

As shown in Fig. 5(b), our method obtains a moderate performance at sizes smaller than 20; on the contrary, in other cases, our method has a clear advantage on urban. This is caused by the diversity of anomaly targets in this scene, and global spectrum is more conducive to discovering the differences between anomaly and background in the condition of the less bands. Why is this so, the ONR, TRC_OC_FDPC, and HCBH are the methods that mainly consider the global spectrum; when the size is small, these methods are more likely to select the involved different anomalies bands from global spectrum, and therefore higher AUC is obtained at the beginning. However, as the number of bands increases, the

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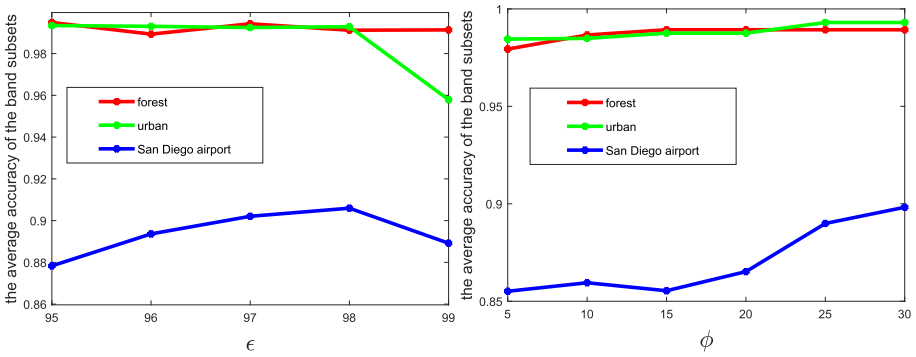


Fig. 4. Determination of parameters ε and ϕ. Left: ϕ varied from 5 to 30 at intervals of 5 in the fixing ε as 96. Right: ε varied from 95 to 99 at intervals of 1 in the fixing ϕ as 25.

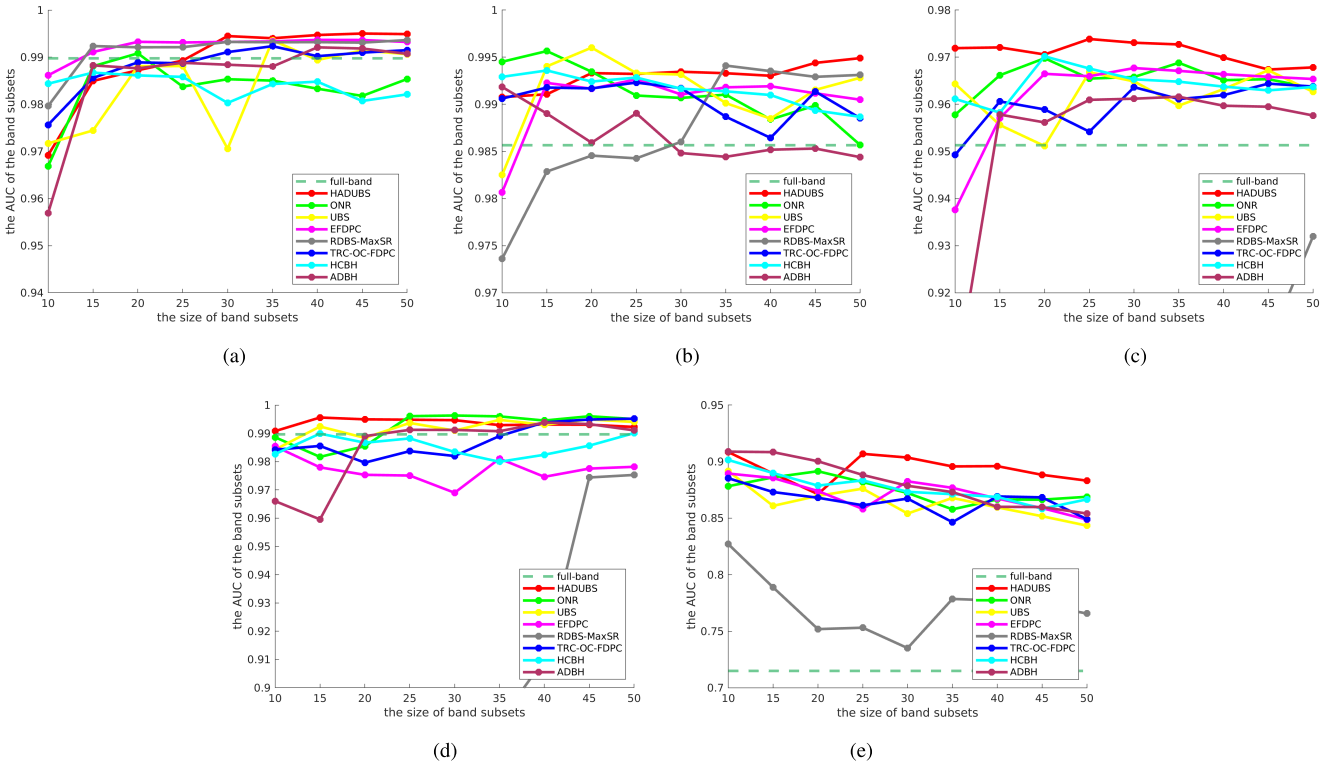


Fig. 5. Comparative experiment of different UBS methods in multiple datasets. (a) Forest. (b) Urban. (c) Urban3. (d) Hyperion. (e) San Diego airport.

bands selected by HADUBS consider the multiple types of anomalies without involving the global spectrum, but rather several consecutive band intervals. As a result, higher AUCs were obtained and there was a trend toward an increase.

For urban3, HADUBS achieves optimal performance with large gaps in any size, shown in Fig. 5(c). Although there is a complex background environment in urban3, the anomaly targets are located mainly in the upper left corner of the figure, the corresponding background, which consists mainly of water, is simpler and is rarely affected by noise, and HADUBS is better suited for scenarios in this situation.

As shown in Fig. 5(d), our method has a more obvious advantage when the anomaly size is smaller in the Hyperion dataset, and it is not difficult to imagine that this is due to the fact that there are few anomaly pixels in the Hyperion; the HADUBS focuses on picking out the few bands that are sensitive to the anomaly targets and suppressed to the background. As a result, when the number of bands is increased, there is more possibility to be affected by the interfering pixels.

The results of San Diego airport are illustrated in Fig. 5(e). Compared with the comparison method, HADUBS achieves a significant advantage at sizes greater than or equal to 25. However, there is no advantage in the first band subsets. It is easy to note that the airport has similar situation as the urban. The different abundances of targets make global spectral information more important in the condition of less bands.

It may be that the band subset equal to 25 fulfills the need for a base scenario for the HAD mission.

To better observe the superiority of HADUBS, the correspondent average AUC of nine band sets is displayed in Table II. It should be noted that the bolded red is for the

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TABLE II
AVERAGE AUC OF EACH UBS METHOD ON MULTIPLE DATASETS, WHERE BOLDED RED ARE FOR HIGHEST ACCURACY, BLACK BOLD FOR SECOND BEST, BOLD AND UNDERLINED FOR THIRD BEST

	forest	urban	urban3	Hyperion	San Diego airport	Average AUC
HADBS	**0.98931**	**0.99304**	**0.97103**	**0.99132**	**0.89363**	**0.96767**
ONR	0.98337	0.98923	**0.96529**	**0.99220**	0.87438	**0.96089**
UBS	0.98422	**0.99133**	0.96173	**0.99183**	0.86388	0.95860
EFDPC	**0.99231**	0.99041	0.96217	0.97712	0.87126	0.95865
RDBSMaxSR	**0.99140**	0.98722	0.87499	0.88793	0.77234	0.93538
TRC_OC_FDPC	0.98833	0.99034	0.95976	0.98757	0.86536	0.95827
HCBH	0.98392	**0.99153**	**0.96418**	0.98544	**0.87685**	**0.96038**
ADBH	0.98585	0.98666	0.95277	0.98511	**0.88126**	0.95833

TABLE III
BAND INDEX FOR DIFFERENT METHODS ON FOREST AND URBAN

	forest_index	urban_index
HADUBS	[133,135,134]	[48,49,37]
ONR	[29, 70, 141]	[20, 78, 143]
UBS	[1 85 169]	[1 88 175]
EFDPC	[141,110,66]	[16,115,145]
RDBSMaxSR	[122,123,127]	[175, 49, 169]
TRC-OC-FDPC	[66,110,141]	[16,115,26]
HCBH	[33,60,168]	[51,94,112]
ADBH	[1,56,83]	[1,55,99]

TABLE IV
COMPARISON OF THE ORIGINAL FEATURE SUBSPACE AND HADUBS

	ours	original feature space
forest	0.98931	0.96703
urban	0.99304	0.96732
urban3	0.97103	0.96994
Hyperion	0.99132	0.94211
San Diego airport	0.89363	0.83652

highest accuracy, bolded black for second best, and bold and underlined for third best. In the forest scenario, EPFDC and RDBSMaxSR get the best average AUC in order, with HADUBS coming in just behind. Although our method is better than most methods only after 25 bands in the urban scenario, it gives the optimal average AUC. In urban3, since our method performs best on an arbitrary subset of bands, it is reasonable to have the best AUC after averaging. In the scene with the least number of anomalies in Hyperion, HADUBS got the third AUC, and this is probably caused by the number of bands increasing; HADUBS is affected by more interfering pixels due to its principle of matching anomaly pixels in our method. In San Diego airport, our method obtained the best average AUC accuracy, improving the accuracy by 0.01237 compared with the second-ranked ADBH method.

Subsequently, an AUC metric for evaluating the overall method was obtained by averaging the AUC across the datasets. HADUBS obtained the optimal detection performance compared with other comparison methods.

To further qualitatively verify the validity of the selected subset of HADUBS bands, we attempted to conduct experiments on visualizing the bands on the forest and urban datasets. We simply selected three bands, and the indexes of the bands for all the methods are shown in Table III. The corresponding bands are visualized in Figs. 6 and 7.

Based on visual bands, it is easy to find that the bands between 122 and 169 can better suppress the background and highlight anomaly targets compared with the other bands. The three bands chosen for HADUBS and RDBSMaxSR fall exactly within this interval, so this could provide some evidence for the validity of HADUBS. However, the bands selected for the RDBSMaxSR do not highlight the targets in the middle two columns. This shows that our method can select the bands that better highlight anomaly targets. The same situation exists for the urban dataset, where only three of our selected bands have better target highlighting. This conclusion is consistent with our analysis of Fig. 8.

To more intuitively observe the salient capabilities of the selected bands, the corresponding anomaly background box plot is also shown in Fig. 8, HADUBS obtained the highest anomaly boxes in both the datasets, which represents the strongest anomaly saliency in our selected bands.

E. Ablation Study

This section conducts an ablation study to evaluate the effectiveness and robustness of the OOD-based feature subspace representation.

First, we replaced the combination of the anomaly subspace and background subspace as the feature space of the original HSI, and the anomaly subspace was replaced with raw HSI. The experimental records obtained in the five scenarios with other experimental conditions held constant are presented in Table IV. Compared with the original feature subspace, the band subset selected from the background subspace and the anomaly subspace combination one always have better AUC, which can prove the effectiveness of our proposed anomaly representation method.

Second, to further validate the robustness of the anomaly feature space derived from representation methods, we take the original feature space and anomaly feature space as inputs of the existing five detectors, respectively. The AUCs of the experiment results are listed in Table V.

It is not difficult to observe that the performance of the version of the anomaly feature space is optimal in most cases metrically, which can prove that the representation method is effective in separating the OOD from the original feature space

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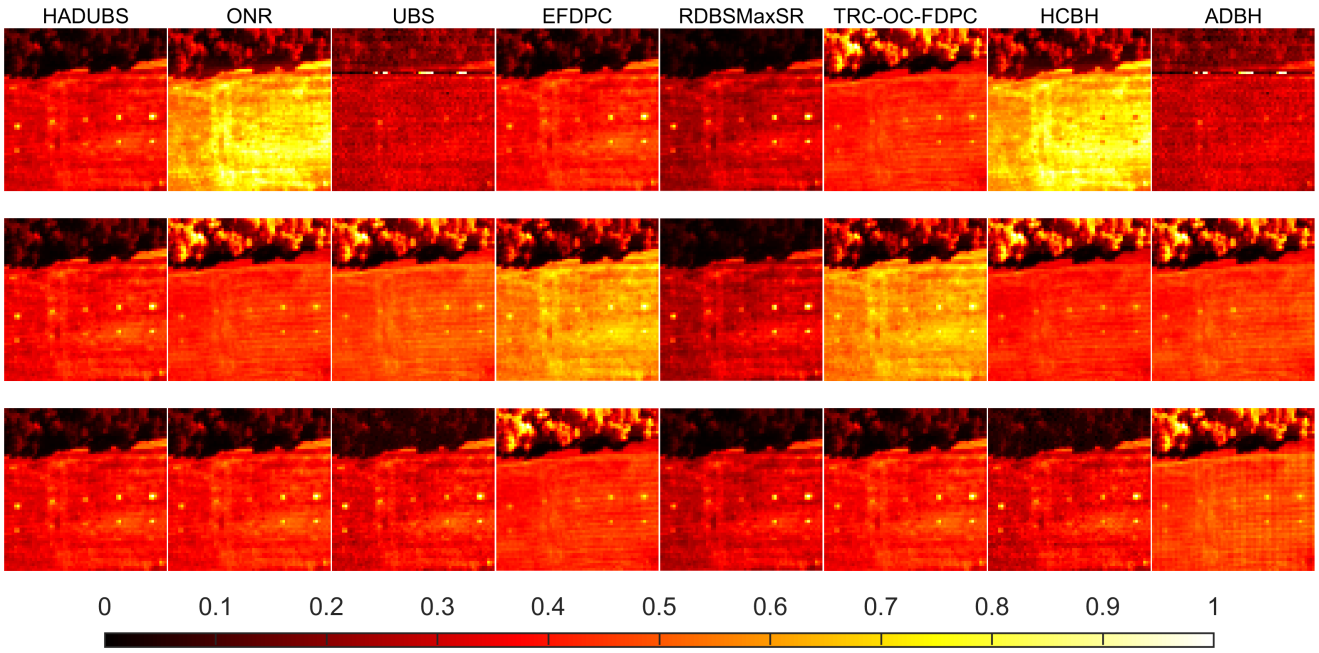


Fig. 6. Corresponding visualization bands in forest.

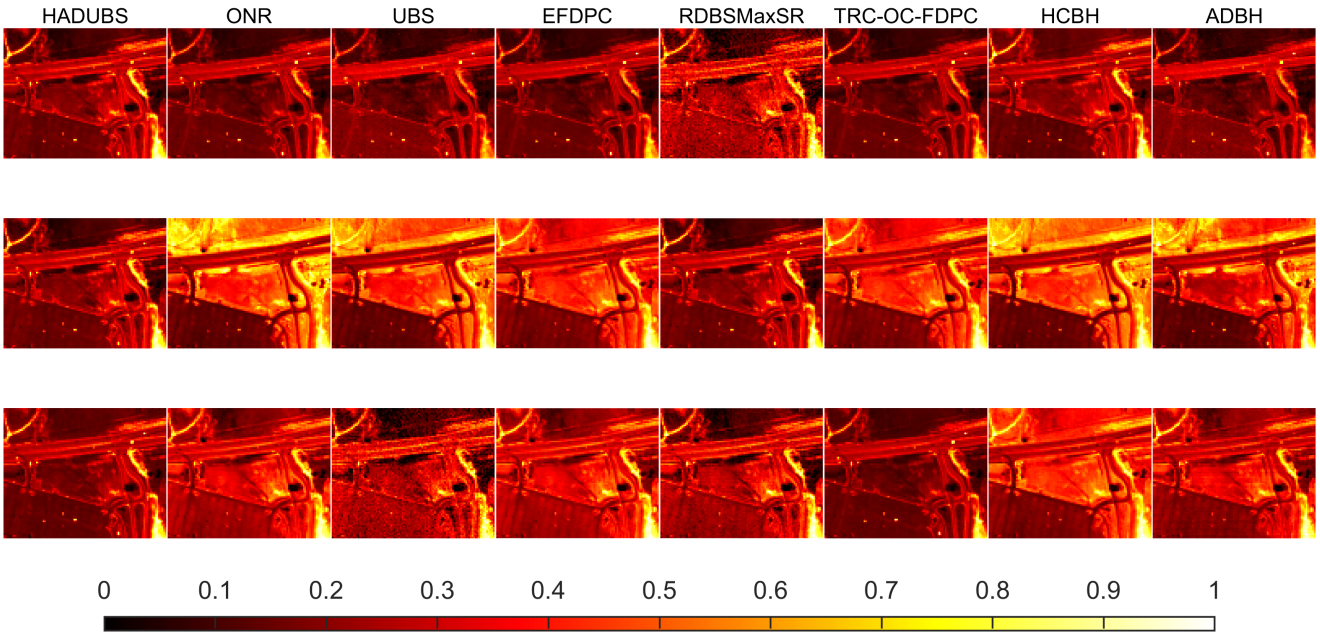


Fig. 7. Corresponding visualization bands in forest.

TABLE V
DETECTION EXPERIMENTS FOR REPRESENTATION METHODS, WHERE O STANDS FOR OOD-BASED FEATURE SUBSPACE REPRESENTATION METHOD

	RX	O-RX	RPCA-RX	O-RPCA-RX	LRASR	O-LRASR	LSMAD	O-LSMAD	LSDM-MoG-RX	O-LSDM-MoG-RX
forest	0.98976	0.99587	0.99351	0.99477	0.94991	0.98634	0.98743	0.99095	0.99104	0.99191
urban	0.98567	0.99536	0.98425	0.99584	0.97336	0.98722	0.99053	0.98907	0.99104	0.97248
urban3	0.95133	0.96529	0.95771	0.97460	0.93620	0.96202	0.96281	0.98160	0.95089	0.95731
Hyperion	0.98966	0.98990	0.98564	0.97646	0.98736	0.96144	0.98813	0.97308	0.97039	0.98197
San Diego airport	0.71504	0.86533	0.78190	0.85901	0.76190	0.84109	0.86298	0.86675	0.73886	0.85329

to some extent. It should be noted that the LSMAD has not get a boost in two scenes. This is probably caused by its empirical parameter selection method representing a limitation.

F. Discussion

In the existing HAD-oriented UBS methods, most methods need to determine the pixel whether or not an anomaly is

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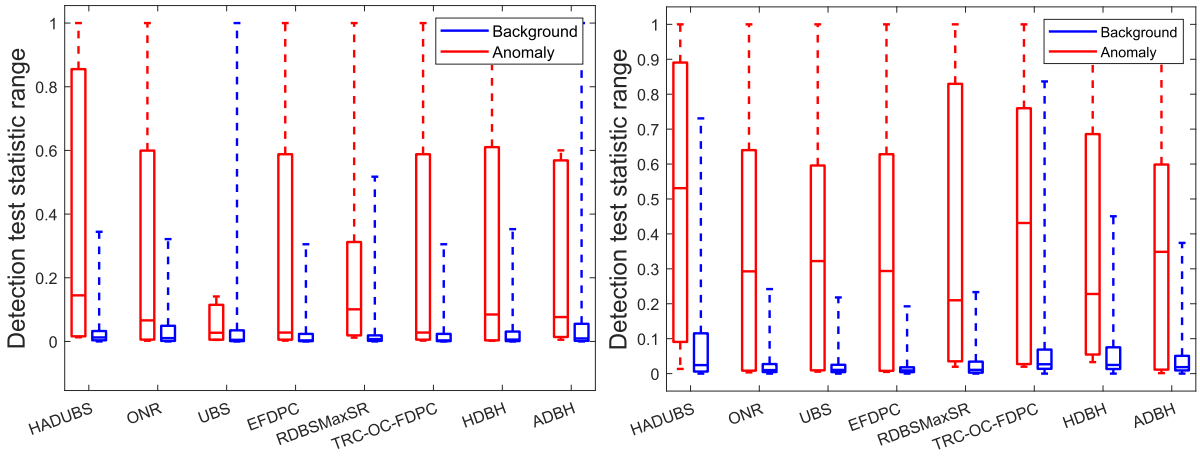


Fig. 8. Anomaly background box plot in forest (left) and urban (right), taking three bands as an example.

TABLE VI
COMPARISON OF RUNNING TIME BY SELECTING THREE BANDS AS AN EXAMPLE IN SECONDS ON THE FOREST DATASET

	HADUBS	ONR	UBS	EFDPC	RDBSMaxSR	TRC-OC-FDPC	HCBH	ADBH
forest	102.9248	0.3447	N\A	0.0188	32.8076	0.0881	0.0485	0.0634

in advance. However, the HAD mission is known to be a challenging task due to extreme pixel distribution. As a result, depending on the existing HAD methods is unreliable for HAD-oriented UBS methods. Based on the observation with respect to the OOD of pixels can reveal the differences and similarities of anomaly representation ability of different bands. Accordingly, we try to propose novel thinking that finds the latent feature space that obeys the OOD term of pixels to better express the anomaly representation ability among bands. In the experiment, some results have given us some inspiration, which will facilitate our further work in this regard.

On one hand, unlike the classification-oriented UBS methods, the uniform band indexing tends to produce better classification results [21], ours and the RDBSMaxSR specially designed for AD choose the relatively continuous bands instead, and achieve better detection performance. On the other hand, our method is able to select the band that best highlights anomalies; however, it is not always optimal for AD tasks. As shown in Table II and Fig. 6, although the selected band from EFDPC is not as good at characterizing anomaly compared with ours, a higher average AUC was obtained. It indicated that the band subset with excellent detection performance is not only related to characterizing anomalies but also affected by other factors, such as the anomaly proportion, the environment in which the anomaly is located, anomaly diversity, and the abundance of abnormal targets, which occur in the analysis within Section IV-D. This is worth further consideration in the future.

G. Stability and Time Analysis

Since the representation method in HADUBS requires repeated iterations, the selected band subsets may be different in each run. In view of this, we ran individually five times in five datasets. In each time, we took the average AUC of nine band subsets as the result of the current run and obtained the mean and standard deviation for plotting in Fig. 9.

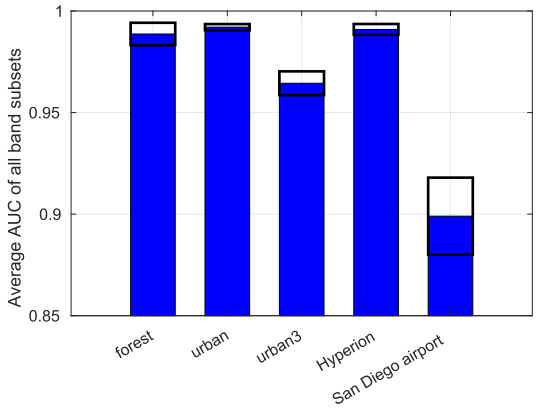


Fig. 9. Stability experiments with HADUBS.

With the exception of the San Diego airport scenario, the performance of HADUBS floats within a very small range of three decimal places in other real-world datasets, which validates the stability of our approach. As for the San Diego airport, this may be caused by the complexity of the simulated scenario, which makes it difficult to separate the anomaly subspace from the original space. As a result, HADUBS causes instability in the inference of anomaly reference diagram in such a complex scenario. Note that despite the presence of AUC jitter, its overall AUC is still higher than that of the other methods, even in the worst case (0.899*(1-0.0189) = 0.882).

The computational burden is an important issue for the HADUBS task. Therefore, we perform a runtime comparison on forest datasets, as listed in Table VI. We found that except for our method and RDBSMaxSR, all the other methods were completed in 1-s time. However, they are all cluster-based

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methods or searching-based, which means that they have to be rerun if the number of acquired bands changes. RDBSMaxSR, HADUBS, and EFDPC are different in that they are based on ranking methods, which can assign priority to all bands in one run. Even though our method took the longest to run, it is within the waiting time range.

V. Conclusion

In this article, we proposed HADUBS, a novel UBS method for HAD tasks, which uses the anomaly subspace and the local sparsity of anomalies to select the band subset in favor of the HAD task. HADUBS mainly contains two key steps, i.e., a representation method that focuses on inferring the OOD and an MI based on LOF that focuses on grasping the local sparsity of anomalies among bands. The massive experiment on the five common datasets has fully validated the robustness and effectiveness of our proposed HADUBS. Our study introduces new thinking on integrating OOD into HAD-oriented UBS, and it effectively avoids the reliance on detectors. However, there are still some limitations such as the time burden being excessive and the redundancies of bands yet to be considered. In future work, we will devote to the above limitations. Besides, the band selection method can play a key role in the hyperspectral object detection task, which can improve the computational efficiency without sacrificing the detection performance. Compared with applying the band selection method on the anomaly detection task, hyperspectral object detection can use more spectral and spatial prior information. Therefore, we will focus more on introducing band selection to the hyperspectral object detection task.

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